

Jenkins and Zabbix Implementation Report

1. What I Implemented

Install Jenkins on ubuntu. Create a simple pipeline job. Pull code from GitHub. Implemented a Jenkins CI/CD setup for automated build and deployment tasks and installed. Build simple Node.js. Run a basic test script. Add one stage for deployment a Zabbix monitoring server on Ubuntu. Added the local system as a monitored host and configured CPU and memory monitoring. Created a CPU utilization alert trigger for monitoring high CPU usage.

2. Challenges Faced

- Configuring Zabbix server and agent and ensuring the host was properly monitored.
- login into zabbix failed while put db scheme zabbix although it need to be empty
- Creating and testing CPU utilization triggers.
- Generating sufficient CPU load to validate trigger behavior.
- Understanding Jenkins pipeline configuration and plugin requirements.
- Adding ip address in zabbix

3. Key Learning

- Learned how to install and configure Zabbix Server and Agent on Ubuntu.
- Learned to monitor CPU and memory metrics using Zabbix templates.
- Learned how Zabbix triggers and alerts work.
- Learned the basics of Jenkins automation and CI/CD concepts.
- Learned zabbix host and trigger
- Learned how to install jenkins in linux and windows
- Learned how to use git in jenkins and built in node
- Understood the importance of integrating monitoring with automation tools.